

3D Stacked FET (3DSFET) Logic and SRAM Technology featuring Single Diffusion Break (SDB) and Back Side Interconnect (BSI) at 48 nm CPP for Advanced Mobile and High Performance Computing (HPC) Applications

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Abstract

For the first time, we report experimental demonstration of highly scalable 3D stacked FET (3DSFET) logic with single diffusion break (SDB) and SRAM bit-cell at 48 nm CPP. Key features include nFET gate-all-around multi-bridge channel (GAA MBCFETTM) vertically stacked on pFET GAA MBCFETTM for reducing logic standard cell height by > 50 %, SDB for reducing logic block area without pitch scaling by > 10 %, self-aligned direct backside contact (SA-DBC) and back side interconnects (BSI) for power and signal lines with relaxed metal pitch, compared to our 3nm node. Moreover, it can provide > 55 % smaller SRAM bit-cell with read static noise margin (SNMR) & write margin (WRM) of 180 & 250 mV, 60 & 50 mV for $V_{DD} = 0.7$ and 0.3 V, respectively. At the same time, SRAM bit-cell leakage decreases by 15.4% at 125°C with unique structural benefits to fully suppress leakage current such as sub-nanosheet, n-p isolation, GIDL and junctions.

Keywords: 3DSFET, SDB, SA-DBC, BSI, SRAM, GAA

Introduction

In order to sustain recently witnessed and accelerated growth trajectory of semiconductor market such as AI, it is essential to satisfy power-performance-area (PPA) for logic and SRAM cost-effectively. However, as physical dimensions approach below 10 nm, critical challenges such as short channel effects, contact resistance, effective device width and metal resistance inevitably necessitate innovative device architecture [1]. One of the most promising candidates is 3DSFET (a.k.a. CFET). It enables to aggressively scale standard cell height by stacking nFET on pFET, likewise SRAM bit-cell by pull-down (PD), pass-gate (PG) on top of pull-up (PU) with making use of SA-DBC & BSI [2-7]. Whereas, SDB can decrease cell-to-cell isolation space without expensive pitch scaling. Thus, 3DSFET with SDB can cost-effectively reduce logic and SRAM area.

Fabrication

Process sequence is summarized in **figs. 2a**, same as our previous reports [2] except additional processes for SDB and shared contacts of SRAM bit-cells. A schematic of 3DSFET SRAM bit-cell is illustrated in **figs. 2b**. While SDB is formed self-aligned to dummy poly-silicon layer prior to MOL contact (**figs. 3a**), shared contacts are processed after flipping a wafer to mitigate routing congestion (**figs. 3b,c**). **Fig. 4** shows a TEM image of monolithic 3DSFET SRAM bit-cell at 48 nm CPP.

3DSFET Logic Technology

A. Leakage Current & External Resistance:

A 3DSFET requires a tall via uniquely to connect top and bottom devices for functional symmetry of both sides of a wafer, which introduces challenges in complex MOL contact & interconnect routing [3]. In addition, during the thinning process of wafer backside, residual Si may cause abnormal leakage current of bottom pFET. However, with optimized tall via and wafer thinning processes, leakage current can be successfully suppressed for both top nFET and bottom pFET (**figs. 5**). Moreover, channel Si thickness can be independently controlled to optimize nFET for the balance between off- and on-current without sacrificing pFET, or vice versa (**figs. 6a**).

Since S/D region of bottom pFET is connected through SA-DBC dissimilar to conventional FinFET [2], it is essential to optimize both eSiGe epi and SA-DBC processes for reducing external resistance (**figs. 6b**).

B. Single Diffusion Break (SDB):

Fig. 7 shows measured $I_{DS}-V_{GS}$ curves of 3D stacked nFET and pFET with SDB for $V_{DS} = 0.05$ & 0.7 V, respectively. This result indicates excellent switching characteristics with sub- V_T swing (SS) of 76/64 mV/dec, DIBL of 66/74 mV/V, and V_T of 0.38/0.20 V for n/pFET. **Figs. 8** compare V_T of SDB and DDB for 3D stacked n/pFET. While there is a negligible V_T difference between SDB and DDB for top nFET, bottom pFET with SDB exhibits ~100 mV smaller V_T than DDB. In addition, channel resistance of the SDB pFET is smaller (not shown), mainly due to localized mechanical stress exerted from SDB.

SDB needs to verify two electric isolation characteristics; (1) between top nFET and bottom pFETs, and (2) between the nearest neighboring 3DSFET. **Figs. 9a,b** show measured isolation characteristics as a function of applied voltage. This implies that both isolation leakage currents are completely suppressed despite the smaller isolation space than DDB and exhibit a very weak temperature dependence (**figs. 9c**).

3DSFET SRAM Bit-Cell Technology

A. Read Static Noise Margin (SNMR) & Write Margin (WRM):

Fig. 10 shows measured butterfly curves of 3DSFET SRAM bit-cell for V_{DD} ranging from 0.3 to 1.0 V. SNMR is 180 mV with a cell current (I_{CELL}) of 17.2 μ A/cell and a cell ratio, $\beta = I_{PD}/I_{PG}$, of ~1.0 at $V_{DD} = 0.7$ V. It increases with V_{DD} and then saturated to ~190 mV for V_{DD} larger than 0.8 V due to disturb current through PG transistor, as shown in **figs. 11a**. It is worth noting that it drops to 60 mV at $V_{DD} = 0.3$ V due to high V_T of PD transistor (not optimized yet) and can be further improved by optimizing V_T of PD and current driving of PG transistor.

WRM is 250 mV with a pull-up ratio, $\gamma = I_{PG}/I_{PU}$, of 2.1 at $V_{DD} = 0.7$ V (**figs. 11b**). It monotonically improves with V_{DD} due to stronger V_{DD} dependence of current driving of PG than PU. However, it quickly drops to 50 mV at $V_{DD} = 0.3$ V due to decreased γ with higher V_T of PG than PU transistor (not optimized yet). To achieve better SNMR and WRM at lower V_{DD} operation, it is essential to minimize SRAM variability and at the same time incorporate read/write assist circuits such as word-line under-drive, negative bit-line voltage, and so on.

B. Leakage Current:

With its unique structural benefits, 3DSFET SRAM bit-cell can completely suppress leakage current such as sub-nanosheet, n-p isolation, GIDL and junctions of PD, PU and 3x PG, as illustrated in **figs. 12**. Although sub- V_T leakage ($I_{S,OFF}$) remains a significant component, this advantage is particularly noticeable at high temperatures, reducing SRAM hold current (I_{HOLD}) by 15.4% at 125 °C, compared to GAA MBCFETTM.

Summary

We reported highly scalable 3DSFET logic with SDB and SRAM bit-cell with stable read & write operation for $V_{DD} = 0.7$ V and down to 0.3 V with 15.4% lower standby current at 125°C, based on 48 nm CPP, SA-DBC and BSI.

References

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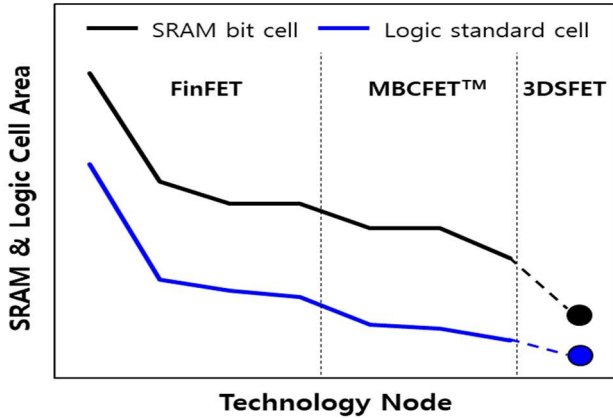
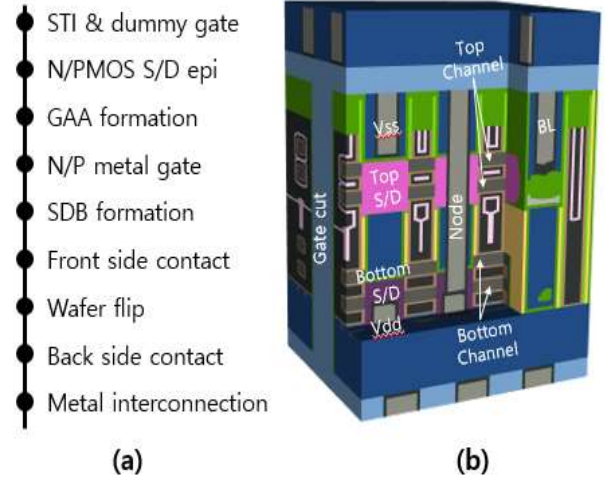
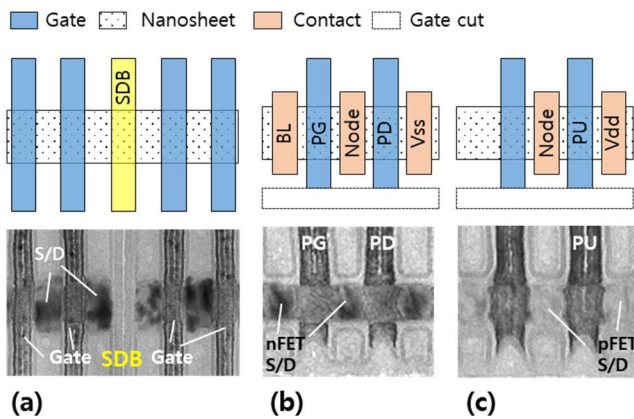


Fig. 1. Scaling trend of SRAM bit-cell and logic standard cell area. 3DSFET with SA-DBC & BSI can reduce logic standard cell height by >50% and SRAM bit-cell by >55%, compared to our 3nm node. Additionally, logic block area can be reduced by >10% with SDB.



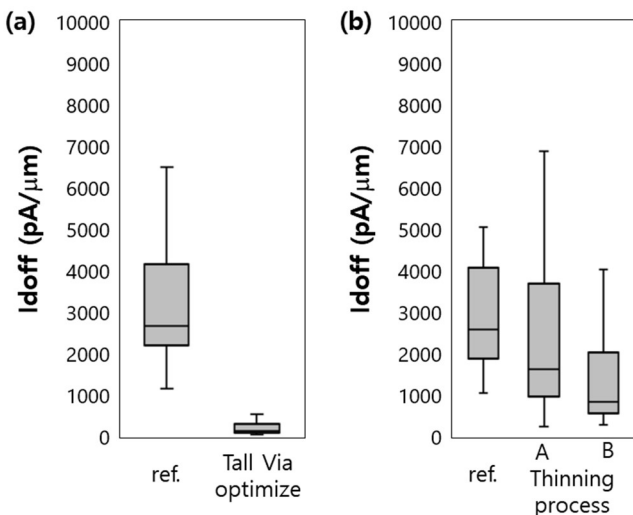
Figs. 2. (a) Process sequence of 3D stacked FET (3DSFET) including SDB and SRAM, and (b) a schematic of 3DSFET SRAM bit-cell.



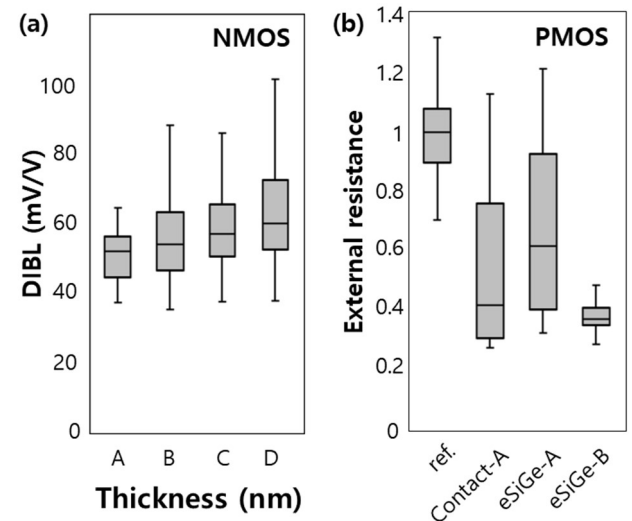
Figs. 3. Top-view schematics and TEM images of (a) single-diffusion break (SDB) (b) SRAM bit-cell (front-side), (c) SRAM bit-cell (back-side).



Fig. 4. Cross-sectional TEM image of monolithic 3DSFET SRAM bit-cell at 48 nm CPP.



Figs. 5. Comparison of leakage current of (a) nFET and (b) pFET for various 3DSFET processes.



Figs. 6. Comparison of (a) DIBL for channel silicon thicknesses of 3D stacked nFET, and (b) external resistance for contact and eSiGe epitaxy processes of 3D stacked pFET

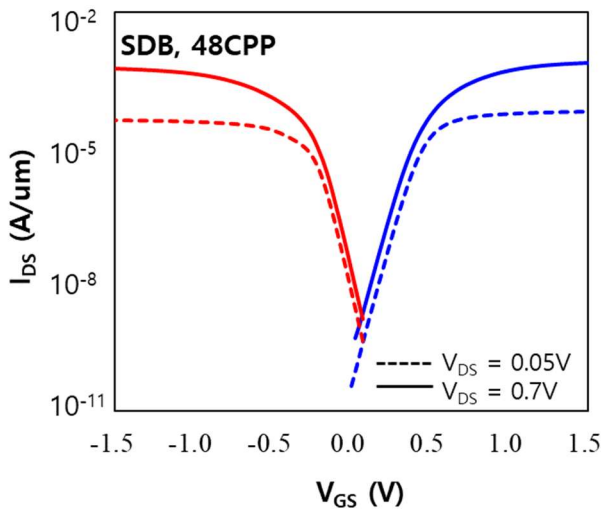
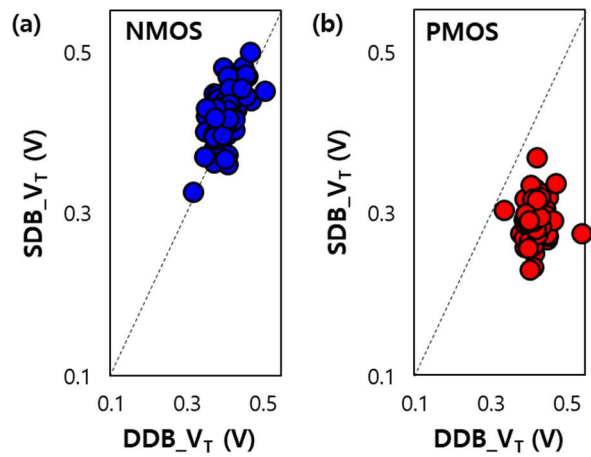
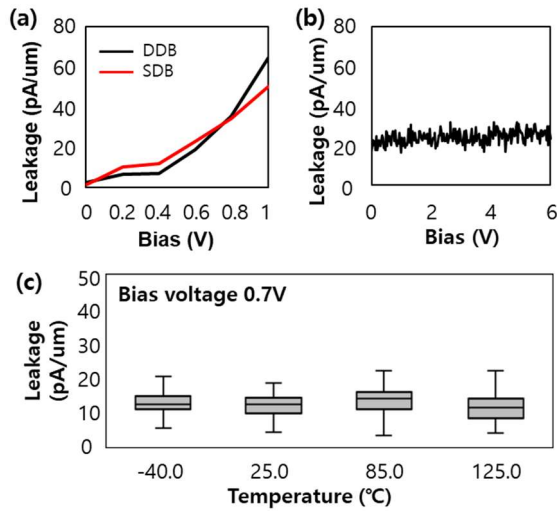


Fig. 7. Measured I_{DS} - V_{GS} curves of 3DSFET with SDB and 2/2 nano-sheets at 48CPP for $V_{DS} = 0.05V$ & $0.7V$, respectively. It shows excellent switching characteristics with SS of 76/64 mV/dec, DIBL of 66/74 mV/V and V_T of 0.38/0.20V for n-/pFET, respectively



Figs. 8. Comparison of V_T of single- (SDB) vs. double- diffusion break (DDB) devices for 3D stacked (a) nFET and (b) pFET. While there is a negligible V_T difference between SDB and DDB for top nFET, bottom pFET with SDB exhibits ~ 100 mV smaller V_T than DDB.



Figs. 9. (a) Comparison of vertical isolation leakage current as a function of applied voltage between top nFET and bottom pFET for SDB and DDB (b) SDB isolation leakage current between nearest neighboring 3DSFET (c) temperature dependence of SDB leakage.

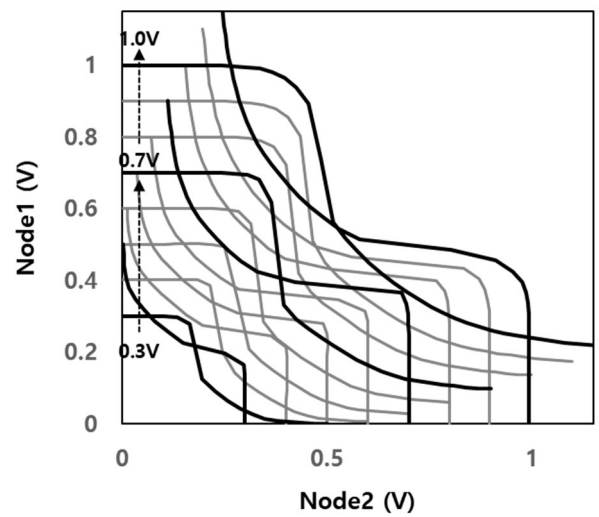
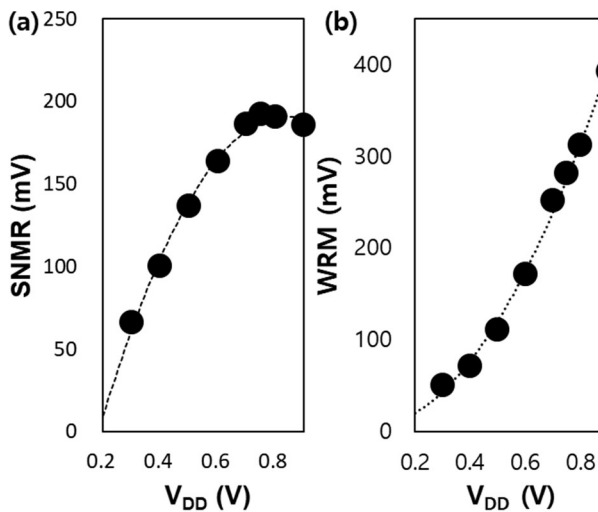
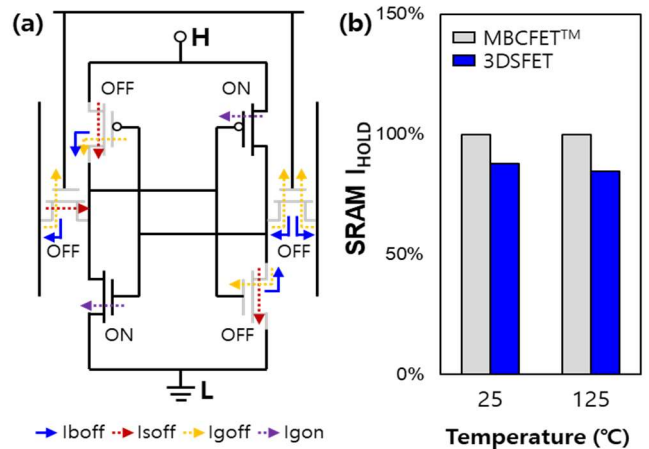


Fig. 10. Measured butterfly curves (SNMR) of 3DSFET SRAM bit-cell for V_{DD} ranging from 0.3V to 1.0V with 0.1V step. SNMR is 180 mV with a cell current of $17.2 \mu A/cell$ for $V_{DD}=0.7V$.



Figs. 11. (a) Read static noise margin (SNMR) and (b) write margin (WRM) for 3DSFET SRAM bit-cell as a function of V_{DD} . While WRM monotonically improves with V_{DD} , SNMR increases with V_{DD} and then saturates to ~ 190 mV for V_{DD} of > 0.8 V.



Figs. 12. (a) Reduced leakage currents (blue) in 3DSFET SRAM bit-cell; sub-nanosheet, n-p isolation, GIDL and junction of PD, PU and PG. (b) Comparison of SRAM hold current (I_{HOLD}). I_{HOLD} can be reduced by 15.4% at 125°C for 3DSFET SRAM bit cell.